UNIVERSITY of KIGALI "Unequalled Education Excellence"

CONTINUOUS ASSESSMENT TEST

GRADUATE SCHOOL

PROGRAMME: MSC-INFORMATION TECHNOLOGY

SESSION: WEEKEND

MODULE CODE AND TITLE: MIT91205: BIG DATA ANALYTICS	**LEVEL AND TRIMESTER:** LEVEL 9, TRIMESTER 3
DATE: 19/07/2026	**TIME:** 9:30AM-11:30AM **SESSION:** WEEKEND
DURATION: 2 HOURS	**TOTAL MARKS:** 100 • TO BE SCALED TO 30 MARKS
EXAMINER/ LECTURER: DR. MUSABE Jean Bosco & DR. BUGINGO EMMANUEL	**MODERATOR:** Dr Wilson Musoni

INSTRUCTIONS TO THE CANDIDATE

1. Questions must be attempted in the answer booklet provided.
2. All queries should be directed to the invigilator; do not communicate or attempt to communicate with any other candidate.
3. You have **TWO HOURS** to complete this paper. You are not allowed to leave the examination room within the **first hour** and in the last 15 minutes of this examination.
4. This is a **CLOSED BOOK** examination.
5. Answer 3 Questions; Question 1 in Section A is compulsory and choose any two (2) other questions in Section B.
6. Please, do not write on the question paper.

SECTION A (COMPULSORY QUESTION) 50 MARKS

QUESTION ONE (Read the paper below and answer the questions that follow):

DataWave Solutions is a multinational consulting firm that specializes in helping organizations transition to big data platforms. They have recently been approached by RetailMax, a traditional retail chain operating 2,000 stores across 30 countries. RetailMax generates massive amounts of data daily, including fifteen million point-of-sale transactions, five million customer loyalty interactions, five hundred thousand social media mentions, and fifty terabytes of store surveillance footage per day. Their current IT infrastructure, which relies on traditional relational databases and on-premises servers, is struggling to handle this data volume, velocity, and variety effectively. The company's IT director, Sarah Mwangi, has identified several critical challenges that are hampering the organization's ability to compete effectively in the modern retail landscape. Monthly sales reports currently take three weeks to generate, severely limiting the company's ability to respond to market trends. The customer analytics team cannot process social media data at all, meaning valuable insights about customer sentiment and brand perception are being lost. Video footage from store surveillance is deleted after seven days due to storage constraints, eliminating any possibility of long-term analysis for security or operational improvement purposes. Real-time inventory tracking is impossible with current systems, leading to stockouts and lost sales. Furthermore, data quality issues across the organization are costing the company approximately fifty million dollars annually through misallocated resources, incorrect pricing, and poor customer targeting.

Required:

a. Assume you have been appointed as the Lead Big Data Architect for RetailMax to design and implement a scalable Big Data platform capable of managing the company's rapidly growing transactional, customer, social media, inventory, and IoT data. Develop a suitable Hadoop ecosystem architecture by selecting and integrating at least six (6) Hadoop ecosystem components (e.g., HDFS, YARN, MapReduce, Hive, Pig, Spark, HBase, Kafka, Sqoop, Flume, Oozie, ZooKeeper). For each selected component, explain its practical role within the proposed architecture, justify why it is appropriate for RetailMax's operational requirements, and demonstrate how these components would interact to support efficient data ingestion, storage, processing, analytics, and decision-making.

15marks

b. RetailMax is planning to establish an enterprise-wide Data Lake that consolidates data from multiple sources, including point-of-sale transactions, e-commerce platforms, mobile applications, supplier databases, IoT sensors, customer feedback, and social media. As a Big Data Governance

Consultant, design a comprehensive Data Governance Framework for the proposed Data Lake. Your framework should include policies, standards, procedures, and technologies for managing data quality, metadata, master data, security, privacy, regulatory compliance, access control, and lifecycle management. Critically justify why each governance component is essential for ensuring trustworthy, secure, and high-quality data, and evaluate how an effective governance strategy would enable RetailMax to maximize business value, support advanced analytics, minimize operational risks, and comply with relevant data protection regulations. **(15 Marks)**

c. Critically evaluate how the Five Vs of Big Data, Volume, Velocity, Variety, Veracity, and Value, are reflected within RetailMax's data ecosystem. For each dimension, provide an appropriate quantitative or qualitative metric, explain its operational and strategic significance, and justify why each characteristic presents both opportunities and challenges for managing large-scale retail data. Furthermore, assess how effectively addressing each of the Five Vs can improve business intelligence, customer experience, operational efficiency, and organizational competitiveness. **(10 Marks)**

d. Judgmentally compare and contrast Traditional Business Intelligence (BI) systems with modern Big Data architectures in the context of RetailMax's business environment. Your discussion should identify the major architectural, technological, and analytical limitations of traditional BI systems that are constraining RetailMax's ability to process large and diverse datasets. Justify why migrating to a modern Big Data ecosystem (such as Hadoop, Spark, NoSQL databases, and cloud-based analytics platforms) would enhance scalability, real-time analytics, predictive decision-making, customer insights, and competitive advantage. Support your arguments with relevant examples from the case study. **(10 Marks)**

SECTION B (CHOOSE ONLY 2 QUESTIONS)

QUESTION TWO **25 Marks**

GlobalPay Solutions is a fintech company that processes payment transactions for e-commerce platforms across one hundred and fifty countries. They handle over one hundred million transactions daily, with peak volumes reaching ten thousand transactions per second. The system must provide strong consistency to maintain accurate account balances, high availability to ensure 99.999 percent uptime, and partition tolerance to remain resilient during network failures. These requirements represent the classic distributed systems trade-off that the CAP theorem describes. Recently, GlobalPay experienced a network partition between their data centers in Singapore and London, which had severe

consequences for the business. During this incident, some users experienced inconsistent account balances where transactions appeared to have been processed but then disappeared, while other users could not access their accounts at all due to system timeouts. The resulting damage was substantial: fifteen million dollars in customer compensation claims, a twenty-five percent drop in user trust scores as measured by post-incident surveys, regulatory investigations from three financial authorities, and approximately five million dollars in lost revenue during the outage. The CTO, Dr. James Okonkwo, has asked you to analyze this incident thoroughly and recommend architectural improvements that would prevent similar incidents in the future.

Required:

a. GlobalPay, a multinational digital payment company, recently experienced a major system outage after one of its distributed data centers became temporarily disconnected due to a network partition. During the incident, some customers were able to complete transactions while others received outdated account balances or were unable to access the payment platform. As a Distributed Systems Consultant, investigate the incident by applying the CAP Theorem. Demonstrate how the system's behavior reflects the three guarantees of Consistency, Availability, and Partition Tolerance (CAP), and explain why a distributed database cannot simultaneously guarantee all three properties during a network partition. Based on your analysis, recommend the most appropriate CAP trade-off for a global financial payment platform such as GlobalPay, providing clear technical and business justifications for your recommendation. **(10 Marks)**

b. Confer the trade-offs GlobalPay faced during the network partition, explaining whether they should have prioritized Consistency or Availability and justifying your answer with specific business considerations. Your response should consider the financial services context and weigh the relative importance of accurate balances versus continuous access for different types of stakeholders. **(8 Marks)**

c. Intricate on the consistency models available in distributed systems, including strong consistency, eventual consistency, and causal consistency, explaining which model you would recommend for GlobalPay's transaction processing and why. Your response should demonstrate an understanding of when each model is appropriate and how the choice impacts system design and user experience. **(7 Marks)**

QUESTION THREE **25MARKS**

HealthInsight is a healthcare analytics company that processes patient data from five hundred hospitals across multiple states. Their daily batch processing workload includes aggregating patient admission statistics, computing readmission rates by hospital and diagnosis, analyzing treatment outcomes across different patient populations, and generating regulatory compliance reports for various government agencies. They currently process fifty terabytes of data daily using a complex SQL-based system running on a single high-performance server, but processing takes twelve hours, which is no longer acceptable for their growing business needs. The Data Engineering Lead, Dr. Amina Hassan, has concluded that the organization needs to migrate to Hadoop's MapReduce framework to achieve the scalability and performance required for their expanding analytics workload. The migration must address the current performance bottlenecks while providing a foundation for future growth and more sophisticated analytics. Dr. Hassan has asked you to design their MapReduce jobs and optimize the batch processing pipeline to meet their evolving requirements.

Required

a. HealthInsight, a national healthcare analytics organization, is planning to implement a Hadoop-based Big Data platform to process millions of electronic health records (EHRs), laboratory results, medical images, wearable sensor data, and hospital admission records collected from hospitals across the country. As a Big Data Engineer, design a MapReduce solution for aggregating and analyzing patient data to generate regional disease statistics, identify high-risk patients, and support healthcare decision-making. Clearly demonstrate how the Mapper, Combiner, Partitioner, and Reducer phases would process the healthcare datasets, explaining the flow of intermediate data between each phase. Justify why each component is essential for improving parallel processing, reducing network traffic, balancing workload across computing nodes, and efficiently processing large-scale healthcare data within the Hadoop ecosystem. **(15 Marks)**

b. Designate the Shuffle and Sort phase in detail, explaining why this is the most expensive part of MapReduce and identifying how HealthInsight can optimize this phase for their specific workload. Your response should address the technical aspects of data movement and sorting, and provide practical optimization strategies. **(10 Marks)**

QUESTION FOUR **25MARKS**

MediaStream is a global video streaming platform with two hundred million subscribers that generates enormous data volumes daily. The platform receives one petabyte of new video content uploaded daily, generates five hundred terabytes of user watch logs, produces one hundred terabytes of user metadata, and creates fifty terabytes of recommendation algorithm output data every day. The company implemented HDFS version 3.x for their data lake, but they are now experiencing significant performance issues that are impacting their analytics capabilities and operational costs. The NameNode memory utilization is consistently at ninety-five percent capacity, threatening system stability. Small file storage is proving highly inefficient, with millions of small metadata files consuming disproportionate NameNode resources. Replication costs have reached three million dollars monthly, representing a substantial and growing operational expense. Query performance for analytics workloads is increasingly slow, frustrating the data science team and delaying business decisions. The VP of Engineering, Maria Santos, has requested a comprehensive review of their HDFS architecture and optimization strategies to address these issues while maintaining the scale and reliability that the platform requires.

Required:

a. Design the HDFS master-slave architecture, explaining the roles of NameNode, DataNode, and Secondary NameNode in MediaStream's deployment. Your response should provide a clear understanding of how these components interact and why the architecture is designed the way it is, including the specific responsibilities of each node type. **(10 Marks)**

b. Describe the impact of storing millions of small files in HDFS, analyzing why this practice is problematic for MediaStream's NameNode memory and overall system performance. Your response should address the specific technical challenges that arise from a large number of small files and provide a quantitative understanding of how this affects NameNode resource consumption. **(15 Marks)**

QUESTION FIVE **25MARKS**

RetailDirect is an e-commerce company with fifty million customers and five hundred thousand products. They have been using MapReduce for two years to process their data, but their data analysts are increasingly frustrated with writing complex Java jobs for what should be straightforward analytical queries. The Head of Analytics, Ms. Sarah Chen, has convinced the executive team to implement Hive for their data warehouse to enable SQL-like queries and accelerate the analytics workflow. Ms. Chen

has established clear requirements for the new system. Analysts must be able to query sales, customer, and product data easily without requiring specialized programming skills. Daily reports must run in under two hours to support time-sensitive business decisions. Historical data spanning five years must be readily accessible for trend analysis and reporting. The system must handle complex joins and aggregations efficiently as the analysis becomes increasingly sophisticated. Furthermore, user growth is twenty percent month-over-month, requiring a scalable solution that can accommodate increasing query loads. Ms. Chen has specifically requested help with performance optimization, particularly on queries that join sales data with customer demographics and product categories, which are currently the most time-consuming operations.

Required:

a. Debate the trade-offs between Hive and other query engines on Hadoop, including Pig and Spark SQL, explaining when RetailDirect should use each tool in their data processing architecture. Your response should provide practical guidance based on different workload characteristics and user requirements. **(15 Marks)**

b. Enlighten the architecture of Apache Hive, including the roles of Metastore, Driver, Compiler, and HiveServer2, and describe how RetailDirect would use each component in their data warehouse deployment. Your response should demonstrate a comprehensive understanding of how Hive transforms SQL queries into distributed processing jobs. **(10 Marks)**

ALL THE BEST!

CAT past paper July 2026 (questions and answers)

MIT91205 CAT, July 2026: Model Answers

Format: Question 1 compulsory (50 marks), then any 2 of Questions 2 to 5 (25 marks each). 2 hours, closed book.

Question 1 (50 marks): RetailMax big data platform

Key facts: 2,000 stores, 30 countries; per day 15M POS sales (about 175 per second), 5M loyalty events, 500K social posts, 50 TB CCTV video; reports take 3 weeks; video deleted after 7 days; no real-time stock; bad data costs $50M a year.

Sizing: 50 TB video x 3 replicas = 150 TB raw disk per day, so cold video needs erasure coding (about 1.5x) or cheap object storage.

(a) Hadoop ecosystem architecture, at least six components (15 marks)

```
POS, loyalty, social -> Kafka/Flume -> HDFS lake (raw/clean/curated) -> Spark (stream, batch, MLlib)
ERP, supplier DBs    -> Sqoop       ->                                -> Hive -> BI dashboards
                                                                       -> HBase (live stock)
            YARN (resources) | ZooKeeper (coordination) | Oozie (scheduling)
```

Component	Role at RetailMax
Kafka	Streams sales, loyalty and stock events; buffers peaks; enables real-time stock
Flume	Collects social feeds, logs and CCTV metadata into HDFS
Sqoop	Bulk import from ERP and supplier databases
HDFS	Data lake on cheap disks; keeps video beyond 7 days; replication for safety
YARN	Shares cluster CPU and memory between Spark, Hive and MapReduce
Spark	Streaming stock updates, cleaning, joins, demand forecasts, sentiment
Hive	SQL warehouse on partitioned ORC tables; cuts 3-week reports to hours
HBase	Millisecond reads and writes for live stock and customer profiles
Oozie / ZooKeeper	Schedule nightly pipelines; keep master services highly available

- Flow: ingest, store raw, stream to HBase, nightly clean and aggregate, Hive for BI, decisions.

Remember: six components, role plus reason, data flow, case problems.

(b) Data governance framework (15 marks)

- **Roles:** governance council (approves policy), data owners (business heads), stewards (define and fix data), custodians (IT run storage and security).

Area	Rule	Tool
Data quality	Valid ids, price > 0, no duplicates, quarantine bad rows	Spark checks, quality dashboard
Metadata	Every dataset registered with owner, source, sensitivity	Hive Metastore, Apache Atlas
Master data	One id for customer, product, store across systems	MDM golden records
Security	Encryption at rest and in transit, strong login	Kerberos, Ranger, Knox
Privacy	Mask phone, card and CCTV faces; consent	Ranger masking, tokenisation
Compliance	GDPR, PCI DSS, local laws in 30 countries; residency	Ranger audit logs, Atlas lineage
Access control	Role and tag based, least privilege, quarterly review	Ranger, LDAP groups
Lifecycle	Hot, warm, cold tiers; retention and legal hold	HDFS storage policies, purge jobs

- **Why:** quality attacks the $50M loss; lineage builds trust; privacy and compliance avoid fines (up to 4% of turnover under GDPR); lifecycle controls video cost.
- **Value:** trusted data for ML, faster decisions, lower risk.

Remember: roles, eight areas, tools, $50M justification.

(c) The Five Vs (10 marks)

V	RetailMax metric	Opportunity	Challenge
Volume	50 TB/day video, 150 TB raw	Long history, security analysis	Storage cost; video deleted
Velocity	15M POS/day, about 175/s	Real-time stock, promotions	Batch takes weeks
Variety	Tables, JSON, text, video	Full customer view, sentiment	RDBMS cannot store it
Veracity	$50M a year lost to bad data	Accurate pricing and targeting	Duplicates, codes differ by country
Value	Fewer stockouts, reports in hours	Better decisions	Data must lead to action

- Handling the Vs gives faster BI, personal offers, real-time inventory and a competitive edge.

Remember: a number per V, opportunity, challenge.

(d) Traditional BI versus modern big data architecture (10 marks)

Aspect	Traditional BI	Big data platform
Scaling	Scale up, hits a ceiling	Scale out on cheap nodes
Data types	Structured only	All types incl. video, text
Schema	On write, rigid	On read, flexible
Processing	Nightly or monthly batch	Batch plus streaming
Analytics	Descriptive	Predictive and prescriptive
Cost per TB	High, so data deleted	Low, tiered

- **Limits today:** one central database (architecture), no video or text analysis (technology), history only (analytics).
- **Why migrate:** keep months of video, real-time stock, MLlib demand forecasts, social sentiment, reports in hours.
- **Balance:** needs new skills and governance; BI tools stay on top of Hive or Spark SQL.

Remember: comparison table, three limits, case benefits, balance.

Question 2 (25 marks): GlobalPay and the CAP theorem

Key facts: 100M transactions/day, peaks 10K/s, 99.999% uptime (about 5 minutes down a year); Singapore to London partition; $15M compensation, $5M lost revenue, 25% trust drop, 3 regulators.

(a) Investigate with CAP and recommend (10 marks)

- **C:** every read sees the latest write. **A:** every request gets an answer. **P:** works despite network splits.
- **Partition happened:** the Singapore to London link failed; in a global system P is not optional.
- **Balances appeared then vanished:** one side kept accepting writes (chose A), then conflicts were rolled back.
- **Timeouts:** other services waited for the remote side (chose C).
- **Why not all three:** with the link down, London cannot see a Singapore write; it must return an old value (lose C) or refuse (lose A).
- **Recommend CP for the ledger:** consensus database (Raft/Paxos, e.g. Spanner, CockroachDB), 5 replicas in 3 regions, majority commit, home region per account, idempotency keys.
- **AP is fine** for catalogues, notifications and dashboards.
- **PACELC:** ledger is PC/EC, accepting about 150 to 200 ms extra latency for correctness.

Remember: define C, A, P; map symptoms; CP with quorum; PACELC.

(b) Consistency or availability? (8 marks)

- **Decision:** consistency for anything that touches a balance.
- **Cost evidence:** inconsistency cost $15M plus trust and regulators; unavailability cost $5M.
- **Customers and merchants:** a delayed payment beats a confirmed one later reversed.
- **Banks and regulators:** accurate settlement is a legal duty.
- **Fraud:** stale balances allow double spending.
- **Soften the impact:** read-only last balance with timestamp, queue payments as "pending", clear error messages, majority quorum so only the minority side stops.

Remember: choose C, stakeholders, case numbers, degraded modes.

(c) Consistency models (7 marks)

Model	Guarantee	Fits
Strong	Reads always see the latest write	Balances, transfers, bookings
Eventual	Replicas converge later; reads may be stale	Likes, catalogues, feeds
Causal	Cause seen before effect by everyone	Comments, chat, activity history

- **GlobalPay:** strong for the ledger; read-your-writes or causal for a user's history; eventual for analytics and notifications.
- **Impact:** strong adds latency, so use home regions; eventual is fast but unsafe for money.

Remember: three models with examples, strong for money.

Question 3 (25 marks): HealthInsight MapReduce

Key facts: 500 hospitals, 50 TB/day, one SQL server takes 12 hours. 50 TB in 128 MB blocks is about 400,000 map tasks running in parallel.

(a) MapReduce for regional disease statistics and high-risk patients (15 marks)

```
HDFS blocks -> Mapper -> Combiner -> Partitioner -> Shuffle/Sort -> Reducer -> HDFS -> Hive
```

- **Mapper:** parse each admission, drop invalid rows, de-identify, look up region from a cached hospital table; emit key (region, diagnosis, month), value (1, stay_days, died).
- **Combiner:** sums counts on the map node; 1M records with 2,000 keys send 2,000 records. Safe because sum is associative.

- **Partitioner:** hash(region, diagnosis) mod R so each key meets in one reducer and big regions spread out.
- **Shuffle and sort:** groups all values of a key together for the reducer.
- **Reducer:** total cases, average stay (from sum and count), mortality rate per region and disease.
- **High-risk job:** key = patient_id; tag admissions, labs and wearable data; reducer joins them and flags 3+ admissions a year, readmission within 30 days, abnormal labs.

Phase	Benefit
Mapper	Thousands of tasks run where the data is
Combiner	Cuts network traffic
Partitioner	Balances load, avoids hot reducers
Reducer	Small aggregated output for reports

Remember: keys and values, combiner safety, de-identify, risk rules.

(b) Shuffle and sort, and optimisation (10 marks)

- **Steps:** map buffer (100 MB) fills, partition and sort, spill to disk, merge, reducers fetch over HTTP, merge sort, group by key.
- **Expensive because:** disk write and read twice, all-to-all network, n log n sorting, serialization, reducers wait for all maps, skew.
- **Optimise:**
  - combiner for sums and counts;
  - compress map output (Snappy or LZ4);
  - drop unused columns and bad rows in the mapper;
  - map-side join for small hospital and ICD tables;
  - tune sort buffer and reducer count;
  - salt or custom partitioner for hot keys;
  - process only new daily partitions; ORC input.

Remember: buffer, spill, fetch, merge, group; five optimisations.

Question 4 (25 marks): MediaStream HDFS

Key facts: 200M subscribers; 1.65 PB new data per day (about 5 PB raw at 3x); NameNode memory at 95%; millions of small files; replication costs $3M a month.

(a) NameNode, DataNode and Secondary NameNode (10 marks)

```
Client --metadata--> NameNode (namespace + block map in RAM) <--checkpoint--> Secondary NN
   \                     ^ heartbeats (3 s), block reports
    \--data directly--> DataNode  DataNode  DataNode ...
```

- **NameNode:** holds namespace, block locations and permissions in RAM; logs changes (fsimage + edits); detects dead nodes; picks rack-aware locations.
- **DataNode:** stores blocks on local disk; heartbeat every 3 s; serves reads; write pipeline DN1 to DN2 to DN3; checksums.
- **Secondary NameNode:** merges fsimage and edit log for fast restart. **Not a failover node.**
- **Production:** NameNode HA (active and standby via JournalNodes and ZooKeeper) and Federation to split `/video`, `/logs`, `/metadata`.
- **Design reason:** data never passes through the master, so throughput grows with DataNodes.

Remember: roles, Secondary is checkpointing, HA, Federation.

(b) Impact of millions of small files (15 marks)

- **Rule:** each file, directory and block costs about 150 bytes of NameNode RAM.
- **Example:** 100 TB as 1 MB files = 100M files = 200M objects = about 30 GB heap; as 128 MB files only about 245 MB.

Problems:

- NameNode heap full even with free disk.
- Long garbage collection pauses and client timeouts.
- Slow restarts (huge fsimage).
- Millions of RPC calls slow every user.
- One task per file, so start-up time exceeds work time.
- Many disk seeks instead of sequential reads.

Fixes:

- Daily compaction into 256 MB files (Spark `coalesce`, Hive merge).
- Store as ORC, Parquet, SequenceFile or Avro; HAR for old files.
- Per-record metadata in HBase.
- HDFS Federation or Apache Ozone.
- Erasure coding (RS 6+3, about 1.5x) for cold data to cut the $3M replication bill; compact first.

Remember: 150 bytes rule, calculation, effects, fixes.

Question 5 (25 marks): RetailDirect and Hive

Key facts: 50M customers, 500K products, Java MapReduce today; needs SQL for non-programmers, reports under 2 hours, 5 years of history, 20% monthly user growth.

(a) Hive, Pig and Spark SQL (15 marks)

Aspect	Hive	Pig	Spark SQL
Language	SQL	Dataflow script	SQL and DataFrames
Users	Analysts, BI	Data engineers	Engineers, data scientists
Engine	Tez, MapReduce, Spark	MapReduce, Tez	Spark, in memory
Speed	Good batch; LLAP for interactive	Batch only	Fast, multi-step
Best for	Shared warehouse, BI reports	Complex ETL, nested data	Heavy joins, ML, streaming

- **Hive:** warehouse of partitioned ORC tables; daily reports through HiveServer2 meet the SQL need.
- **Spark SQL:** sales x customers x products joins, segmentation, MLlib, same metastore.
- **Pig:** only existing ETL on raw logs; new ETL in Spark.
- **Speed up the slow join:** partition sales by date; bucket sales and customers on customer_id; broadcast the 100 MB products table; ORC with statistics; pre-aggregated daily tables.
- **Growth:** scale out on YARN or cloud; LLAP or Trino for many users.

Remember: comparison, when to use each, join tuning.

(b) Hive architecture (10 marks)

```
BI tools (JDBC/ODBC) -> HiveServer2 -> Driver -> Compiler <-> Metastore
                                    -> Tez/MapReduce/Spark on YARN -> HDFS
```

Component	Role
HiveServer2	Accepts queries from many users, sessions, authentication
Driver	Controls query life cycle: compile, execute, fetch
Compiler	Parses SQL, checks schema, builds and optimises the plan
Metastore	Tables, columns, partitions, locations, statistics in MySQL or PostgreSQL
Execution engine	Runs the task DAG on YARN over HDFS

- **Query path:** BI tool, HiveServer2, Driver, Compiler asks Metastore, optimiser prunes partitions and picks map join, DAG runs on YARN, results return.

Remember: four components, query steps, diagram.